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Sex differences in *semitendinosus* muscle fiber-type composition

Gaspard Fournier[1] | **Clara Bernard**[2] | **Maxime Cievet-Bonfils**[1] | **Raymond Kenney**[3] | **Maxime Pingon**[1] | **Elliot Sappey-Marinier**[1] | **Bénédicte Chazaud**[2] | **Julien Gondin**[2] | **Elvire Servien**[1,4]

[1]Department of Orthopedic Surgery and Sport Medicine, FIFA medical center of excellence, Hôpital de la Croix-Rousse, University Lyon 1, Lyon, France

[2]Institut NeuroMyoGène, CNRS UMR 5310, INSERM U1217, Université Claude Bernard Lyon 1, Univ Lyon, Lyon, France

[3]Department of Orthopaedics, University of Rochester Medical Center, Rochester, New York, USA

[4]EA 7424 - Inter-University Laboratory of Human Movement Science, Univ Lyon, Université Claude Bernard Lyon 1, Villeurbanne, France

Correspondence
Julien Gondin, Institut NeuroMyoGène (INMG), CNRS 5310 – INSERM U1217 - UCBL1, Faculté de Médecine et de Pharmacie, 8 Avenue Rockefeller, 69008 Lyon, France.
Email: julien.gondin@univ-lyon1.fr

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Sex differences in muscle fiber-type composition have been documented in several muscle groups while the hamstring muscle fiber-type composition has been poorly characterized. This study aimed to compare the *semitendinosus* muscle composition between men and women. Biopsy samples were obtained from the *semitendinosus* muscle of twelve men and twelve women during an anterior cruciate ligament reconstruction. SDH and ATPase activities as well as the size and the proportion of muscle fibers expressing myosin heavy chain (MyHC) isoforms were used to compare muscle composition between men and women. The proportion of SDH-positive muscle fibers was significantly lower (37.4 ± 11.2% *vs.* 49.3 ± 10.6%, $p < 0.05$), and the percentage of fast muscle fibers (*i.e.,* based on ATPase activity) was significantly higher (65.8 ± 10.1% *vs.* 54.8 ± 8.3%, $p < 0.05$) in men versus women. Likewise, men muscles exhibited a lower percentage of the area that was occupied by MyHC-I labeling (35.6 ± 10.1% vs. 48.7 ± 8.9%; $p < 0.05$) and a higher percentage of the area that was occupied by MyHC-IIA (38.3 ± 6.7% *vs.* 32.5 ± 6.5%; $p < 0.05$) and MyHC-IIX labeling (26.1 ± 9.6% *vs.* 18.8 ± 8.5%; $p = 0.06$) as compared with women muscles. The cross-sectional area of MyHC-I, MyHC-IIA, and MyHC-IIX muscle fibers was 31%, 43%, and 50% larger in men as compared with women, respectively. We identified sex differences in *semitendinosus* muscle composition as illustrated by a faster phenotype and larger muscle size in men as compared with women. This sexual dimorphism might have functional consequences.

KEYWORDS
hamstrings, muscle biopsies, muscle composition, myosin heavy chain

Gaspard Fournier, Clara Bernard, Julien Gondin, and Elvire Servien authors are equally contributed to this work.

1 | INTRODUCTION

Skeletal muscle is the largest organ in the human body, accounting for up to 40% of body weight, and is a very heterogeneous tissue. One of the main characteristics of skeletal muscle relies on its composition of different types of muscle fibers that can be activated to perform various functional tasks. Human skeletal muscle is composed of three major fiber types that can be identified according to either their ATPase activity or their expression of the myosin heavy chain (MyHC) isoforms.[1] On that basis, muscle fibers are usually classified as slow type 1 (expressing MyHC-I), fast intermediate type 2A (expressing MyHC-IIA), and fast type 2X (expressing MyHC-IIX). Each fiber type possesses specific contractile responses (*i.e.,* slow *vs*. fast) and metabolic activities (*i.e.,* aerobic *vs*. anaerobic) that are required to perform a broad range of functional activities ranging from short-duration and high-force level contractions (*e.g.,* explosive movements) to long-lasting and low-intensity contractions (*e.g.,* endurance exercises).

Sex differences in muscle fiber-type composition have been largely documented in humans.[2] Several studies have demonstrated that men possess a lower proportion of type 1 and a higher proportion of type 2A as compared with women.[3-10] However, these observations mainly arose from analysis of quadriceps muscle biopsy samples[4,7] (*i.e., vastus lateralis*), while only a few investigations focused on other muscle groups such as *biceps brachii*,[10] *lateral gastrocnemius*,[11] or *tibialis anterior*[12] muscles. Surprisingly, fiber-type composition of hamstring muscles (*i.e., semitendinosus*, *semimembranosus*, and *biceps femoris* muscles), which are the main knee flexors and are involved in various explosive tasks such as sprinting and running, have been poorly characterized. So far, hamstring muscle fiber-type composition has been mainly examined in men[13-15] and has never been directly compared between men and women.[16,17] The investigation of potential sex differences in hamstring muscle fiber-type composition is particularly relevant if we consider the higher incidence of hamstring muscle strain injuries (HSI) in men *vs*. women,[18,19] the higher susceptibility to exercise-induced muscle injury in the fast *vs*. slow muscle fibers,[20,21] as well as the higher fatigability observed in men as compared with women.[22]

In the present study, we analyzed biopsy samples from the *semitendinosus* obtained in men and women during an anterior cruciate ligament (ACL) reconstruction. We combined histochemical analyses with MyHC immunostaining in order to characterize oxidative and ATPase activities together with the size and the proportion of muscle fibers expressing MyHC-I, MyHC-IIA, and MyHC-IIX isoforms.

TABLE 1 Patient's characteristics

	Men ($n = 12$)	Women ($n = 12$)	*p* value
Age (years)	34 ± 8	35 ± 8	$p = 0.9$
BMI (kg/m²)	25.5 ± 3.4	24.8 ± 5.3	$p = 0.7$
Tegner activity score	5.9 ± 0.7	5.3 ± 0.6	$p = 0.08$
Delay between ACL injury and surgery (days)	73.4 ± 9.6	79.6 ± 13.1	$p = 0.4$

Note: Data are presented as mean ±SD. ACL: anterior cruciate ligament; BMI: Body mass index.

2 | MATERIALS AND METHODS

2.1 | Population

Study enrollment began in March 2019. Inclusion criteria were as follows: patients scheduled for an ACL reconstruction using *semitendinosus* and *gracilis* autograft and aged between 18 and 50 years old. Exclusion criteria included history of prior hamstring injuries, neuromuscular diseases, or chronic use of corticosteroids. Demographics information is summarized in Table 1. Sport played and activity level were recorded using Tegner Activity Scale[23] before ACL injury (Table 1). The Tegner Activity Scale aims at providing a standardized method of grading work and sporting activities. The score varies from 0 to 10. A score of 0 represents sick leave or disability pension because of knee problems, whereas a score of 10 corresponds to participation in national and international elite competitive sports. A score of 5–6 corresponds to participation in recreational sports. No professional athletes were included. Overall, twelve men and twelve women were included. The delay between the ACL injury and surgery as well as the Tegner activity score was not significantly different between the two groups (Table 1). Gender and age-appropriate consent were obtained under a protocol approved by Hospital's ethics committee (N°19–52) and by data protection committee (19–083) of Lyon.

2.2 | Muscle samples

Muscle samples were obtained from *semitendinosus* of patients undergoing ACL reconstruction using a close-ended tendon stripper (Figure 1). *Semitendinosus* muscle biopsies were frozen in isopentane placed in liquid-nitrogen and kept at −80°C until use. Cryosections (10 µm) were prepared for staining and immunohistochemical analyses (all compounds from Sigma-Aldrich unless indicated).

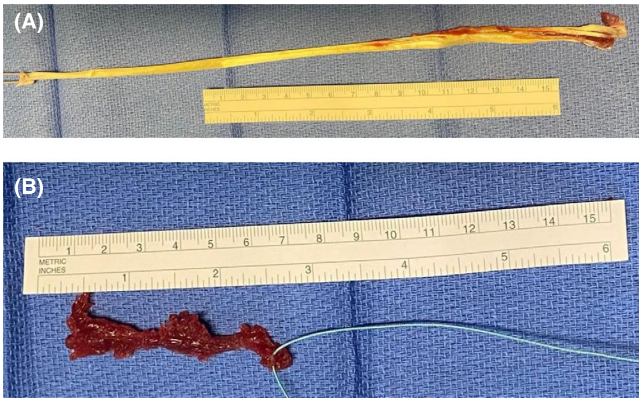

FIGURE 1 (A) Tendon and (B) *semitendinosus* muscle samples obtained during anterior cruciate ligament reconstruction. The panel A illustrates the insertion of the *semitendinosus* muscle on the distal tendon while the panel B shows the amount of muscle tissue obtained during the surgery

2.3 | Succinate dehydrogenase (SDH) staining

Succinate dehydrogenase activity was revealed histochemically as a marker of muscle fiber oxidative capacity. Briefly, cryosections were dried at room temperature for 30 min and further incubated in a solution composed of nitroblue tetrazolium 1.5 mM, EDTA 5 mM, succinic acid 48 mM, sodium azide 0.75 mM, methyl-phenylmethlyl sulfate 30 mM, and phosphate buffered to pH 7.6 for 45 min at 37°C. Cryosections were then washed in deionized water for 5 min, dehydrated in ethanol 50% for 2 min, and mounted for viewing (Eukitt).

2.4 | ATPase staining

ATPase activity was revealed histochemically as a marker of muscle fiber type. Cryosections were incubated in a solution composed of barbital acetate 0.1 M and HCl 0.1 M at pH 4.5–4.6 for 5 min at room temperature. Cryosections were then washed in deionized water and incubated in a solution with ATP, sodium barbital, and calcium chloride for 25 min at room temperature. Then, they were washed 3 times with calcium chloride 1%, were incubated in cobalt chloride 2% for 10 min, washed 5 times with sodium barbital 0.1 M, incubated with ammonium sulfide 2% for 30 s, and finally rinsed with tap water. Cryosections were then dehydrated in ascending alcohols baths (50%, 70%, 80%, 95%, and 100%), cleared with xylene, and mounted with Eukitt mounting medium.

2.5 | MyHC immunostaining

Cryosections were incubated with primary antibodies overnight at 4°C, washed with PBS, and further incubated with secondary antibody for 1 h at 37°C. The following antibodies were used: anti-MyHC-I (mouse IgG2b, BA-D5, 1/100), anti-MyHC-IIA (mouse IgG1, SC-71, 1/100) and anti-MyHC-IIX (mouse IgM, 6H1, 1/100) (all from Developmental Studies Hybridoma Bank), and anti-laminin (rabbit, L9393, 1/200, Sigma-Aldrich). Secondary antibodies were Alexa Fluor 405-conjugated donkey anti-mouse IgG2b (1/200), Alexa Fluor 488-conjugated anti-mouse IgG1 (1/200), Alexa Fluor 546-conjugated donkey anti-mouse IgM (1/200), and Cy-5-conjugated donkey anti-rabbit antibodies (1/200) (all from Jackson Immunoresearch). Slides were washed with PBS and mounted in Fluoromount-G medium.

2.6 | Image analysis

All images were recorded and analyzed in a blinded fashion. For both SDH and ATPase staining, 4 randomized images were recorded from each section with a Axioskop Zeiss microscope at 20X magnification connected to an AxioCam ICc Zeiss camera.

SDH-positive fibers were stained dark blue (high oxidative enzymatic activity) while SDH-negative fibers were stained light blue (low oxidative enzymatic activity). ATPase-positive fibers (sensitive to pH 4.6, *i.e.,* fast muscle fibers) were stained light blue while ATPase-negative fibers (not sensitive to pH 4.6, *i.e.,* slow muscle fibers) were stained dark blue. For both stainings, the number of positive and negative fibers was quantified with Imaging Software (Image J, NIH) and normalized to the total number of fibers (*i.e.,* ~450 fibers per sample).

For MyHC immunostaining, 4 randomized images were recorded from each section with an Imager Z1 Zeiss microscope at 20X magnification connected to a CoolSNAP MYO camera. Using the combination of antibodies described above, MyHC-I was labeled by a blue staining, MyHC-IIA by a green staining, MyHC-IIX by a red staining, and laminin by far red staining that was converted into false white for a clear visible delineation of muscle fibers. Muscle fiber cross-sectional area was quantified for each MyHC isoform with ICY software (version 2.1.4.0; http://icy.bioimageanalysis.com). The percentage of each MyHC was then expressed relative to whole muscle cross-sectional area (*i.e.,* % total area representing ~320 fibers per sample).[7] The number of fibers expressing each MyHC isoform was also quantified with Imaging Software (Image J, NIH) and was normalized to the total number of fibers (*i.e.,* % total number representing ~500 fibers per sample).

2.7 | Statistical analysis

Statistical analysis was performed using GraphPad Prism Software (version 7.00). Data distribution was initially investigated using Kolmogorov-Smirnov test. Student's t test was performed for demographic variables. Student's t test with Welch correction was used to test differences between men and women for other variables. Cohen's d values were calculated with G*Power software (Version 3.1.9.6).[24] Small, moderate, and large effects were considered for $d \geq 0.2$, ≥ 0.5, and ≥ 0.8, respectively. Data are presented as mean ±SD with significance set at $p < 0.05$.

3 | RESULTS

3.1 | SDH staining

SDH staining was used to compare the proportion of muscle fibers showing high oxidative enzymatic activity between men and women (Figure 2A-B). Men had a significantly lower proportion of oxidative muscle fibers (37.4 ± 11.2%) as compared with women (49.3 ± 10.6%; $p < 0.05$; $d = 1.10$; Figure 2C).

3.2 | ATP staining

ATPase staining was performed to compare the proportion of slow and fast muscle fibers between men and women (Figure 2D-E) A significant higher proportion of fast muscle fibers was observed in men (65.8 ± 10.1%) as compared with women (54.8 ± 8.3%, $p < 0.05$, $d = 1.19$; Figure 2F).

3.3 | MyHC immunostaining

MyHC immunostaining was performed to compare the size and the proportion of muscle fibers expressing MyHC-I, MyHC-IIA, and MyHC-IIX isoforms between men and women (Figure 3A-B) The cross-sectional area of MyHC-I, MyHC-IIA, and MyHC-IIX muscle fibers was 31%, 43%, and 50% larger in men as compared with women, respectively ($p < 0.05$; Table 2). Considering the variation in the size of both type 1 and type 2 muscle fibers between men and women, we first quantified the proportion of muscle fibers expressing MyHC-I, MyHC-IIA, and MyHC-IIX isoforms with regard to the area occupied by each MyHC isoform. Men had a significantly lower percentage of MyHC-I positive-muscle fibers (35.6 ± 10.1% vs. 48.7 ± 8.9%; $p < 0.05$; $d = 1.38$; Figure 3C) and a higher percentage of MyHC-IIA (38.3 ± 6.7% *vs.* 32.5 ± 6.5%; $p < 0.05$; $d = 0.87$; Figure 3D) as compared with women. Men also showed a trend toward a higher percentage of MyHC-IIX (26.1 ± 9.6% *vs.* 18.8 ± 8.5%; $p = 0.06$; $d = 0.81$)

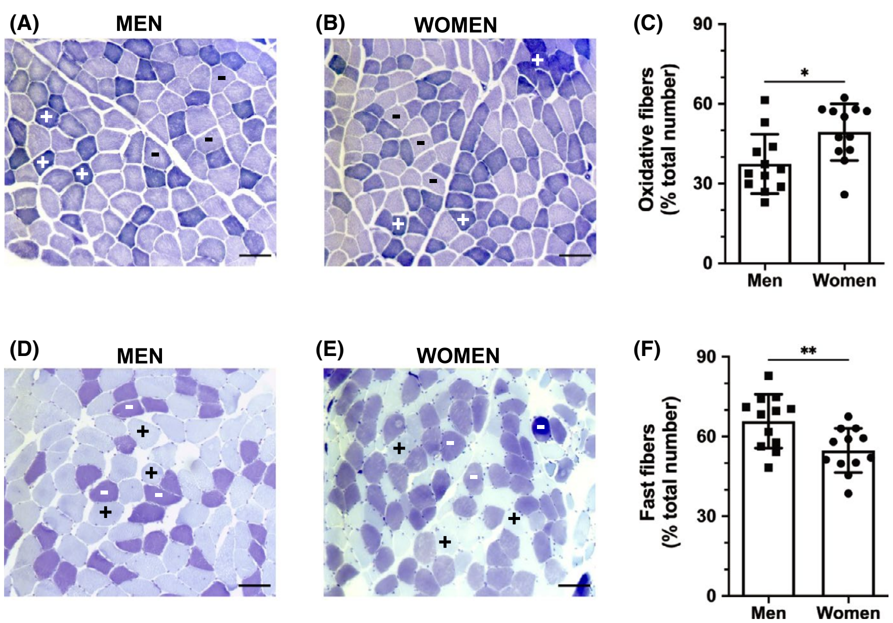

FIGURE 2 (A-B) Representative SDH staining of *semitendinosus* muscle obtained in men (panel A) and women (panel B). SDH-positive fibers are stained dark blue (high oxidative enzymatic activity as shown by symbol "+") while SDH-negative fibers are stained light blue (low oxidative enzymatic activity as shown by symbol "-"). Scale bar: 50 μm. (C) Quantification of the proportion of oxidative fibers in men ($n = 12$) and women ($n = 12$). (D-E) Representative ATPase staining of *semitendinosus* muscle obtained in men (panel D) and women (panel E). ATPase-positive fibers (sensitive to pH 4.6, *ie*, fast muscle fibers) are stained light blue (as shown by symbol "+") while ATPase-negative fibers (not sensitive to pH 4.6, *ie*, slow muscle fibers) are stained dark blue (as shown by symbol "-"). Scale bar: 50 μm. (F) Quantification of the proportion of fast fibers in men ($n = 12$) and women ($n = 12$). Significantly different from men: $^{*}p < 0.05$; $^{**}p < 0.01$. Results are mean ± SD

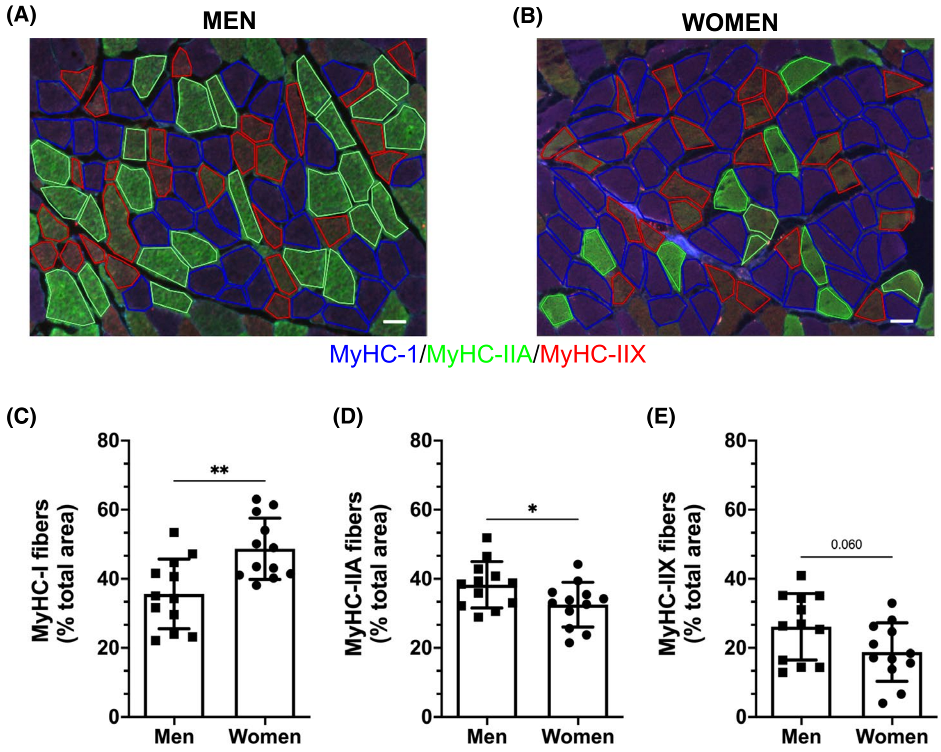

FIGURE 3 (A-B) Representative MyHC immunostaining of *semitendinosus* muscle obtained in men (panel A) and women (panel B). MyHC-I is labeled by a blue staining, MyHC-IIA by a green staining, MyHC-IIX by a red staining, and laminin by far red staining that is converted into false white for a clear visible delineation of muscle fibers. Regions of interest for the quantification of muscle fiber cross-sectional area are drawn. Scale bar: 50 µm. (C-E) Proportion of MyHC-I (panel C), MyHC-IIA (panel D), and MyHC-IIX (panel E) expressed in percentage of the whole muscle cross-sectional area (*ie*, % total area) in men (n = 12) and women (n = 12), respectively. Significantly different from men: $^{*}p < 0.05$; $^{**}p < 0.01$. Results are mean ± SD

TABLE 2 *Semitendinosus* muscle fiber size and composition in men and women

	Men	Women	*D*
Fiber size (µm^2)			
MyHC-I	4905 ± 1873	3396 ± 1053*	0.99
MyHC-IIA	5256 ± 1616	3011 ± 1099***	1.62
MyHC-IIX	4622 ± 2199	2327 ± 1039**	1.33
Fiber-type composition (% total number)			
MyHC-I	32.3 ± 8.2	42.0 ± 8.7*	1.14
MyHC-IIA	33.8 ± 5.7	31.3 ± 8.0	0.36
MyHC-IIX	33.9 ± 7.4	27.4 ± 7.6*	0.87

Note: Cohen's *d* values are reported in the right column. Significantly different from men: $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$. Data are presented as mean ± SD.

expressing muscle fibers as compared with women (Figure 3E).

When expressed as a relative number of fibers, men also had a significantly lower percentage of MyHC-I positive-muscle fibers and higher percentage of MyHC-IIX expressing muscle fibers as compared with women ($p < 0.05$; Table 2). No significant difference was observed for the expression of MyHC-IIA between the two groups (Table 2).

4 | DISCUSSION

In the present study, we combined histochemical and immunostaining analyses to compare hamstring muscle fiber-type composition between adult men and women. The most important finding of this study was the identification of sex differences in muscle composition as illustrated by a faster muscle phenotype and larger muscle fiber size in men as compared with women.

4.1 | *Semitendinosus* muscle fiber-type composition

Human skeletal muscle is a very heterogenous tissue featuring a composition of various muscle fiber types. Recent evidence illustrated that muscle fiber-type composition is dependent on species,[25] anatomical location,[26] and sex.[2] Despite the critical role of hamstring muscles in human locomotion and sprint-based activities, their fiber-type composition has been poorly characterized,[13-17] with most studies being performed with samples obtained mainly *via* autopsy from men[13,14,16] and only a few investigations focusing on the *semitendinosus* muscle.[16,17]

When considering our data obtained in men only, we found that the *semitendinosus* muscle had a larger distribution of fast muscle fibers (*i.e.,* ~66% of ATPase-positive fibers and ~64–68% of MyHC-II) as compared with slow muscle fibers. This fast phenotype associated with the *semitendinosus* muscle is in accordance with that previously observed in ten (including 7 men) cadavers[16] (*i.e.,* ~60% of type 2 fibers) but differs with the balanced fiber-type distribution reported in a previous study of sixteen adult patients (including 14 men) who underwent ACL reconstruction[17] (*i.e.,* 50% of type 2 fibers). Considering that we also analyzed biopsy samples from patients with the same pathology (*i.e.,* ACL reconstruction), this discrepancy is hard to explain. It is noteworthy that the histochemical and immunostaining investigations performed here are more thorough than those performed by Eriksson et al.,[17] our quantitative analyses involving a larger number of fibers (*i.e.,* ~500 fibers *vs*. ~200 fibers, respectively).

Our results obtained in men also underline that the *semitendinosus* muscle displays the fastest phenotype among the hamstrings as illustrated by the higher proportion of fast fibers (> 60%) as compared with that reported in the *biceps femoris* (*i.e.,* ~51–53% of MyHC-II)[14,15] and in the *semimembranosus* (*i.e.,* ~50% of type 2 fibers)[16] muscles.

We showed that men *semitendinosus* muscle exhibited a faster muscle fiber-type composition as compared with women as illustrated by a smaller proportion of SDH-positive fibers (*i.e.,* Figure 2A-C), a higher percentage of ATPase-positive fibers sensitive to specific acid pH (*i.e.,* Figure 2D-F), as well as a lower percentage of area occupied by MyHC-I positive-muscle fibers (*i.e.,* Figure 3C) together with a higher percentage of area occupied by MyHC-IIA (*i.e.,* Figure 3D) and MyHC-IIX (*i.e.,* $p = 0.06$; Figure 3E) positive-muscle fibers. These results are in accordance with the sexual dimorphism previously described in the *quadriceps* muscle (*i.e., vastus lateralis*), with men having a lower proportion of type 1 fibers and higher percentage of type 2 fibers as compared with women. Taken together, muscle fiber-type composition differs between men and women in both *vastus lateralis* and *semitendinosus* muscles, therefore independently of the functional role of the considered muscle (*i.e.,* knee extension *vs*. knee flexion, respectively).

Finally, we demonstrated that the cross-sectional area of MyHC-I, MyHC-IIA, and MyHC-IIX muscle fibers was larger in men as compared with women, the difference being higher for type 2 (+43–50%) than for type 1 (+31%) muscle fibers. These results are similar with those reported in the *vastus lateralis*,[4,7] *tibialis anterior*,[12] and *biceps brachii*[10]muscles. They indicate that sex differences in muscle fiber size are consistently found among human skeletal muscles in both upper and lower limbs and independently of the subject characteristics (*e.g.,* healthy volunteers, patients undergoing ACL reconstruction).

4.2 | Potential consequences of sex differences in *semitendinosus* muscle composition

The functional consequences of the sexual dimorphism in *semitendinosus* muscle fiber-type composition are unclear. On the basis of isolated and single muscle fiber experiments, it was reported that MyHC composition is a key determinant of muscle fiber contractile velocity[27] and fatigue resistance.[28] One could therefore speculate that the faster phenotype in men might be beneficial for rapid/ballistic movements while the slower muscle typology in women might provide an advantage in terms of fatigue resistance. However, it has been recently reported that the *biceps femoris* MyHC distribution was unrelated to any measure of maximal or explosive strength in a cohort of adult healthy men.[15]

Although sex differences in fatigue resistance have been documented, illustrating a greater fatigue resistance in women than in men,[29,30] it should be kept in mind that the etiology of fatigue is multifactorial and involves factors other than muscle typology (*e.g.,* impairment of the central nervous system, alterations in excitation-contraction coupling).[28,31]

Hamstring muscles play a key role in several explosive muscle activities and are frequently injured. Indeed, HSI is the most prevalent non-contact injury in various sports such as soccer[32] and sprinting.[33] Interestingly, evidence of a higher incidence of HSI in men than in women is emerging.[18,19] Considering the higher susceptibility to exercise-induced muscle injury of fast muscle fibers than the slow ones,[20,21] one could speculate that the faster hamstring muscle fiber-type composition in men might be a predisposing factor to HSI. Although the present study does not aim at addressing this issue, a direct link between HSI and fiber-type distribution seems questionable. Indeed, the *semitendinosus* is the fastest muscle among the hamstrings but is also the less injured muscle as it is involved in less than 10% of soccer-related HSI while the biceps femoris accounting for more than 80% of HSI.[34] One could, therefore, assume that factors other than muscle fiber-type composition (*e.g.,* anatomical and/or biomechanical factors[35-37]) may play a key role in the incidence of HSI.

4.3 | Limitations

There are several limitations in our study. Muscle samples were obtained from *semitendinosus* of patients undergoing ACL reconstruction. It remains to demonstrate whether a similar sex difference exists in both healthy subjects and professional athletes. The surgical procedure of ACL reconstruction precluded any functional measurements at the time of the biopsy. In addition, our study focused on the *semitendinosus* muscle which accounts only for ~25% of the total hamstring muscle volume, whereas hamstring functional performance is influenced by both the *biceps femoris* and the *semimembranosus* muscles. Finally, *semitendinosus* muscle was obtained in patients aged from 22 to 49 years in men and from 24 to 49 years in women. Our sample size was too small to consider subgroup of patients according to their age (*e.g.,* <30 years *vs.* 30–40 years *vs.* >40 years). Further studies are warranted to determine the influence of age on *semitendinosus* muscle fiber-type composition in men and women.

5 | CONCLUSION

The present study is the first to compare hamstring muscle fiber-type composition between adult men and women. Thanks to the combination of histochemical and immunostaining analyses, we showed sex differences in *semitendinosus* muscle fiber-type composition as illustrated by the faster phenotype and larger muscle fiber size in men as compared with women.

6 | PERSPECTIVE

The impact of this sexual dimorphism on explosive performance, resistance to fatigue, and/or incidence of injury is still unknown. Further studies are needed to determine whether and to what extent sex differences in *semitendinosus* muscle fiber-type composition influence muscle performance *in vivo*.

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CONFLICT OF INTEREST

GF, CB, MCV, RK, MP, ESM, BC, and JG have nothing to disclose. ES reports non-financial support from Corin, non-financial support from Smith and Nephew, outside the submitted work.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

ORCID

Julien Gondin https://orcid.org/0000-0002-3108-605X

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